

TOPIC 8. CERAMIC MATERIALS

- **Introduction**
- **Structure of Ceramic Materials**
- **Mechanical Properties of Ceramic Materials**
- **Glasses**

INTRODUCTION

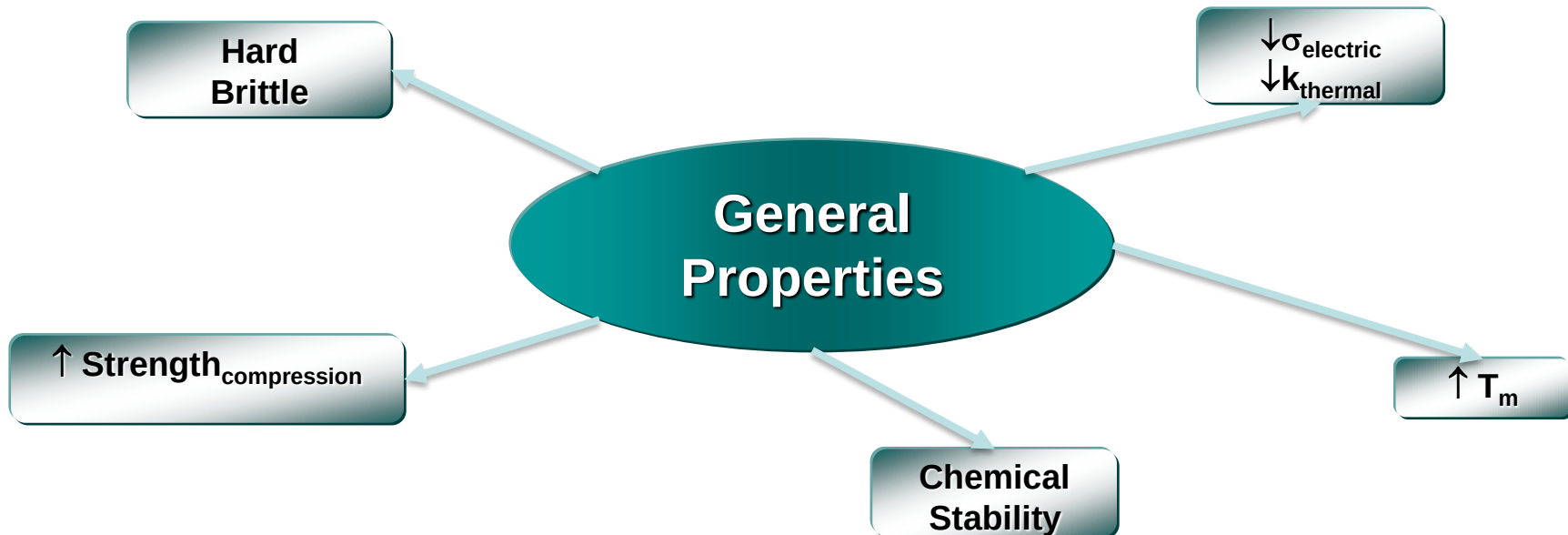
Ceramic materials: inorganic materials composed of metals and non-metals bonded by ionic and/or covalent bonds

Ceramic materials can be **crystalline or amorphous**

Abundant sources for ceramic materials



GENERAL PROPERTIES



GENERAL PROPERTIES

Why ceramic materials have such properties?

- **High Young's Modulus and high melting points**

⇒ Strong bonds (covalent and /or ionic)

- **Hard and brittle and high compressive strength**

⇒ Strong bonds (covalent and /or ionic)

- **Limited electrical and thermal conductivity**

⇒ Absence of electronic cloud

- **Low thermal shock resistance**

⇒ High elastic modulus and low thermal conductivity

- **Refractory**

⇒ Stability at high temperature (NO CREEP)

- **Resistance to oxidation/corrosion**

⇒ Chemical stability

CLASSIFICATION

Glasses

SiO_2 + additives

Traditional Ceramics (clay)

- ⇒ Porous ceramics (bricks, pottery, china)
- ⇒ Compact ceramics (porcelain, earthenware)
- ⇒ Refractory ceramics

Clay: $\text{Al}_2\text{O}_3 \cdot \text{SiO}_2 \cdot \text{H}_2\text{O}$

Silica: SiO_2

Feldspar: $\text{K}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2$

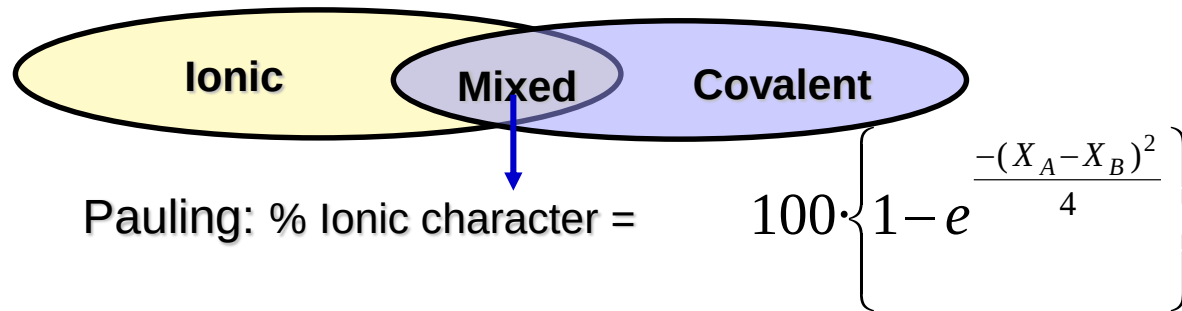
Engineering Ceramics or Advanced Ceramics :

- ⇒ Refractory ceramics (SiC , Al_2O_3 , ZrO_2 , BeO , MgO).
- ⇒ Piezoelectrics and Ferroelectrics: BaTiO_3 , SrTiO_3
- ⇒ Electro-optics: LiNbO_3
- ⇒ Abrasive ceramics: nitrides and carbides Si_3N_4 , SiC
- ⇒ Molecular membranes
- ⇒ Superconducting ceramics ($\text{YBa}_2\text{Cu}_3\text{O}_7$)
- ⇒ Biomaterials : Hydroxyapatite

BONDING AND STRUCTURE IN CERAMICS

Ceramic Bonds

Bonding of ions in ceramic material is either (or both) ionic and covalent



Percentage of ionic and covalent character of the bond for some ceramic materials  determines the **CRYSTALLINE STRUCTURE**

Ceramic Material	Atoms in bond	$X_A - X_B$	% Ionic Character	% Covalent Character
MgO	Mg—O	2,3	73	27
Al ₂ O ₃	Al—O	2,0	63	37
SiO ₂	Si—O	1,7	51	49
Si ₃ N ₄	Si—N	1,2	30	70
SiC	Si-C	0,7	11	89

ELECTRONEUTRALITY IN IONIC CERAMICS

Ionic structure: packing of **anions** with **cations** in interstitials

The ions tend to **pack densely** in order to **reduce E_{total}** . The **type of Bravais lattice** formed depends on:

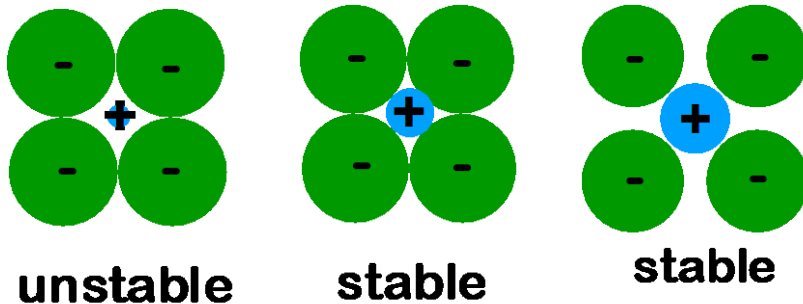
Sizes $C^+ A^- \Rightarrow (r_{\text{cation}} < r_{\text{anion}})$

The relation between radius when A^- and C^+ are in contact $\Rightarrow$ **minimum**

Electroneutrality

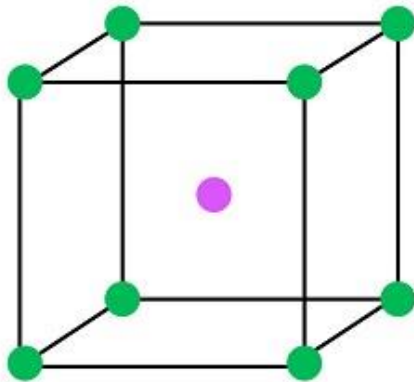
Coordination Index

Increasing C.I $\Rightarrow$ increase stability

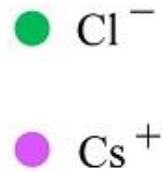


Cation/anion Radius ratio r_C/r_A	Arrangement of A^- around C^+ central and C.I.	structure
0.732-1.0	8 Corners of a cube	CsCl
0.414-0.732	6 Corners of an octahedron	Rock salt (NaCl)
0.225-0.414	4 Corners of a tetrahedron	zincblende (ZnS)
0.155-0.225	3 Corners of a triangle	
<0.155	2	

SIMPLE CUBIC STRUCTURE: CsCl



Unit cell - Cesium Chloride



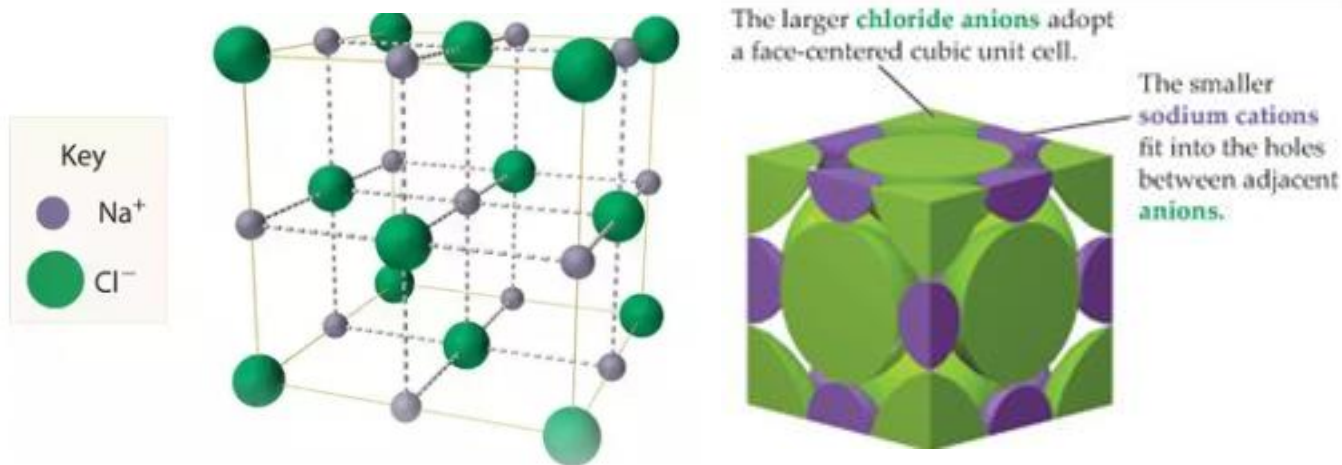
$$\frac{r_{\text{Cs}^+}}{r_{\text{Cl}^-}} = 0,94 > 0,732 \Rightarrow \text{C.I.} = 8 \Rightarrow \text{C.I.: } 8$$

- Cl^- : corners of the cube
- Cs^+ : centre of the cube
- C.I.: 8

Ceramics that have this type of structure **halides**: **CsBr, TlCl, TlBr.**

Note: this is **not a BCC**, two atoms!

ROCK SALT STRUCTURE: NaCl

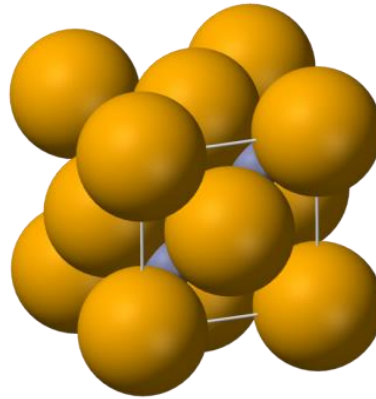
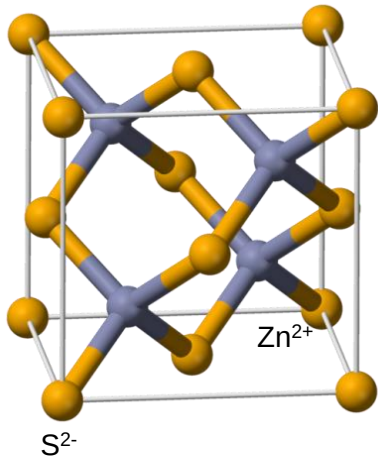


$$\frac{r_{Na^+}}{r_{Cl^-}} = 0,56 > 0,414 \Rightarrow C.I. = 6 \Rightarrow \text{Octahedral coord.}$$

- Cl⁻: FCC lattice points
- Na⁺: all octahedral interstitials.
- 4 Na⁺ and 4 Cl⁻ per unit cell C.I.=6

Ceramics that have this type of structure **oxides: MgO, CaO, FeO, NiO**

ZINCBLLENDE STRUCTURE - ZnS



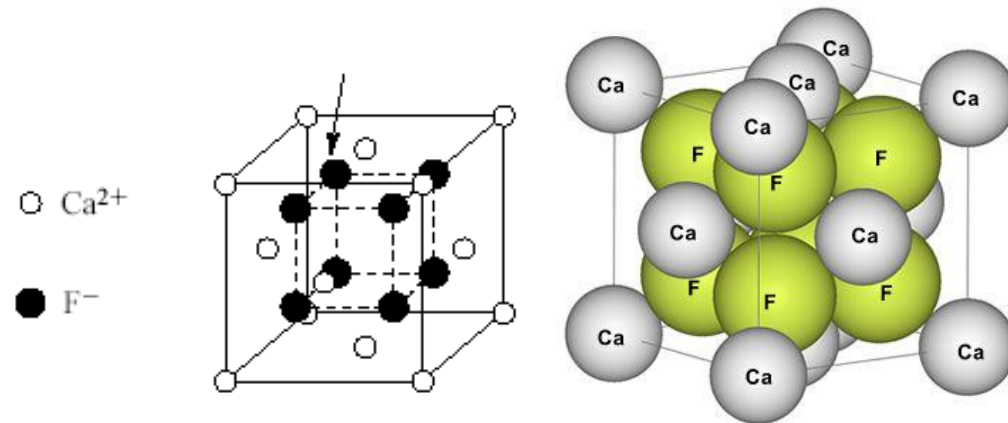
$$\frac{r_{\text{Zn}^{2+}}}{r_{\text{S}^{2-}}} = 0.345 \Rightarrow C.I. = 4$$

- S^{2-} : FCC lattice points
- Zn^{2+} : $\frac{1}{2}$ tetrahedral interstitials
- 4 Zn^{2+} and 4 S^{2-} per unit cell

Ceramics that have this type of structure: **semiconductors : CdS, HgTe, NiAs, SiC, GaAs**

According to Pauling bond **Zn-S ~87% covalent**

FLUORITE STRUCTURE CaF_2



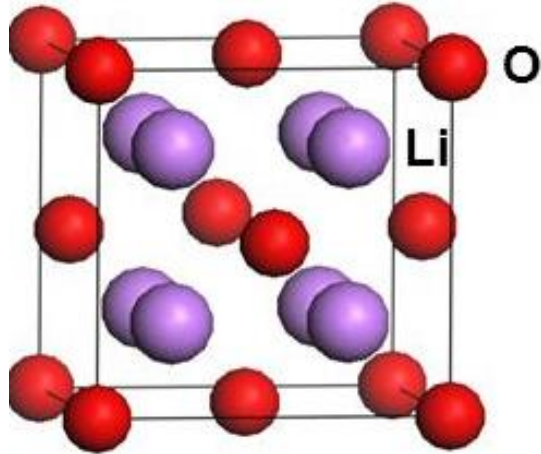
$$\frac{r_{\text{Ca}^{2+}}}{r_{\text{F}^-}} = 0.752 \Rightarrow C.I. = 8$$

C.I. (Ca^{2+}): 8 ; C.I. (F^-): 4

- Ca^{2+} : FCC lattice points
- F^- : all tetrahedral interstitials
- 4 Ca^{2+} and 8 F^- per unit cell

Ceramics that have this type of structure **oxides**: UO_2 , ThO_2 , CeO_2 , YSZ

ANTI-FLUORITE STRUCTURE Li_2O

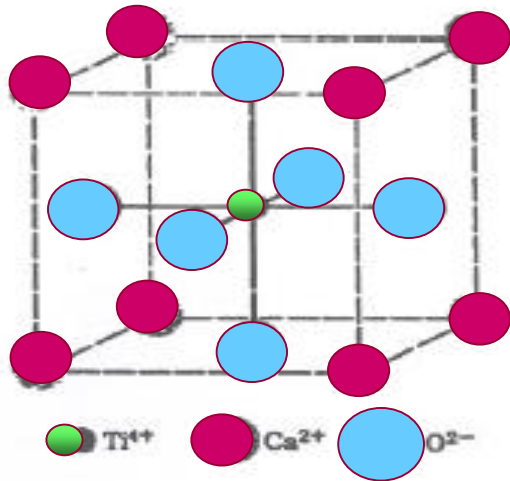


$$\frac{r_{\text{Ca}^{2+}}}{r_{\text{F}^{-}}} = 0.752 \Rightarrow C.I. = 8$$

C.I. (Li^{+}): 4 ; C.I. (O^{2-}): 8

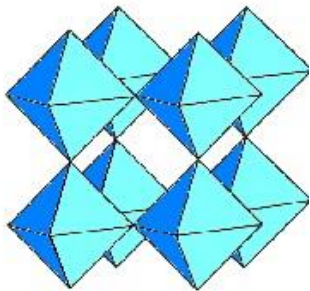
- O^{2-} : FCC lattice points
- Li^{2+} : all tetrahedral interstitials
- 8 Li^{+} and 4 O^{2-} per unit cell

Ceramics that have this type of structure **oxides**: Na_2O , K_2O , Mg_2Si

PEROVSKITE ABO_3 

A and B cations with different size
($r_A \gg r_B$)

- O^{2-} and Ca^{2+} : **fcc** packing:
 - Ca^{2+} in the **corners**
 - O^{2-} in **face centers**
- Ti^{4+} : **1/4 octahedral sites**
- C.I. (Ti^{2+}): 6 ; C.I. (Ca^{2+}): 12
- 3 O^{2-} , 1 Ca^{2+} and 1 Ti^{4+} per unit cell



TiO_6 Octahedra

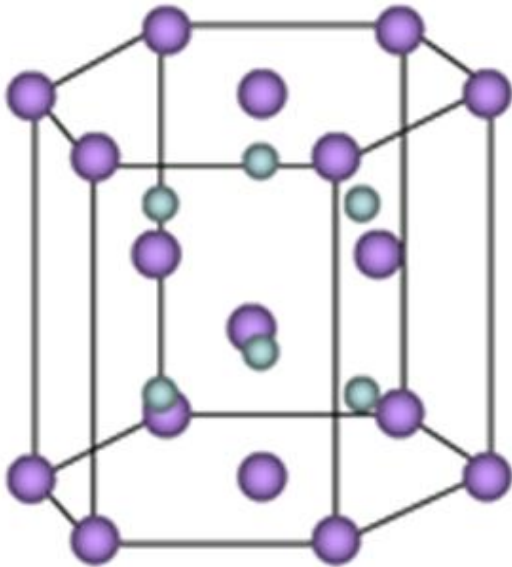
Ceramics that adopt this type structure:

BaTiO_3 , CaTiO_3 , SrTiO_3 , PbZrO_3 , KNbO_3 , LiNbO_3 ,...

Ferroelectric Materials,

Magnetic Superconductor properties ($\text{YBa}_2\text{Cu}_3\text{O}_7$)

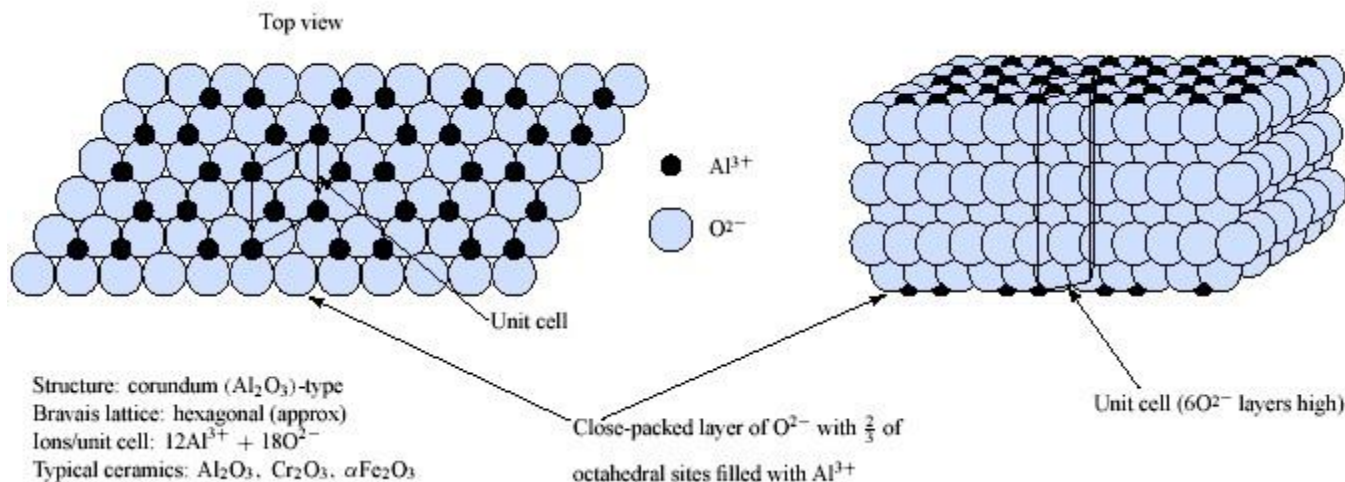
HCP STRUCTURE: CORUNDUM (ALUMINA)



- O^{2-} : HCP lattice points $\rightarrow$ 6 ions
- Al^{3+} : $2/3$ octahedral interstitials $\rightarrow$ 4 ions
- I.C. (Al^{3+}): 6 ; I.C. (O^{2-}): 6
- 6 O^{2-} , 4 Al^{3+} per unit cell

Ceramics that have this type of structure:

Cr_2O_3 , Fe_2O_3 , ...



Summary of Some Common Ceramic Crystal Structures

<i>Structure Name</i>	<i>Structure Type</i>	<i>Anion Packing</i>	<i>Coordination Numbers</i>		<i>Examples</i>
			<i>Cation</i>	<i>Anion</i>	
Rock salt (sodium chloride)	AX	FCC	6	6	NaCl, MgO, FeO
Cesium chloride	AX	Simple cubic	8	8	CsCl
Zinc blende (sphalerite)	AX	FCC	4	4	ZnS, SiC
Fluorite	AX ₂	Simple cubic	8	4	CaF ₂ , UO ₂ , ThO ₂
Perovskite	ABX ₃	FCC	12(A) 6(B)	6	BaTiO ₃ , SrZrO ₃ , SrSnO ₃

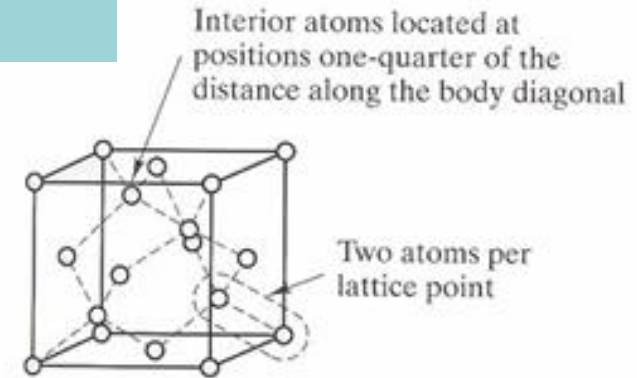
COVALENT CERAMICS

They are **structural** ceramics

DIAMOND → Structure type blend

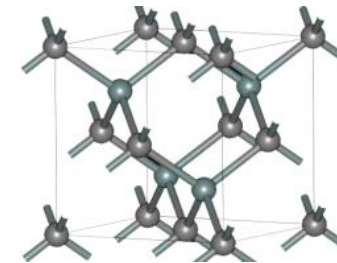
C → sp^3 → c.i. 4 → Tetrahedral. CC_4 Bond 100% covalent.

- ↑ wear resistance ↑ hardness
- ↑ tensile strength Insulator



SiC → Diamond type structure (spherullite)

- Applications: Good abrasive properties. 89% covalent bond
- High hardness, chemically inert.

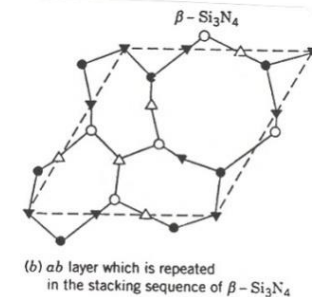
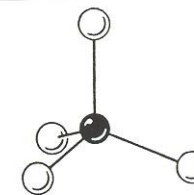


Si_3N_4 → Cutting Elements, blades, rotors

Si → sp^3 → c.i. 4 → SiN_4 Tetrahedra

N → sp^2 → c.i. 3 → N coordinated to 3 Si

Open structure. 70% covalent bond



PROPERTIES OF CERAMICS

Hardness

Very hard solids

Hardness (Covalent) > hardness (Ionic)

(Diamond, SiC)

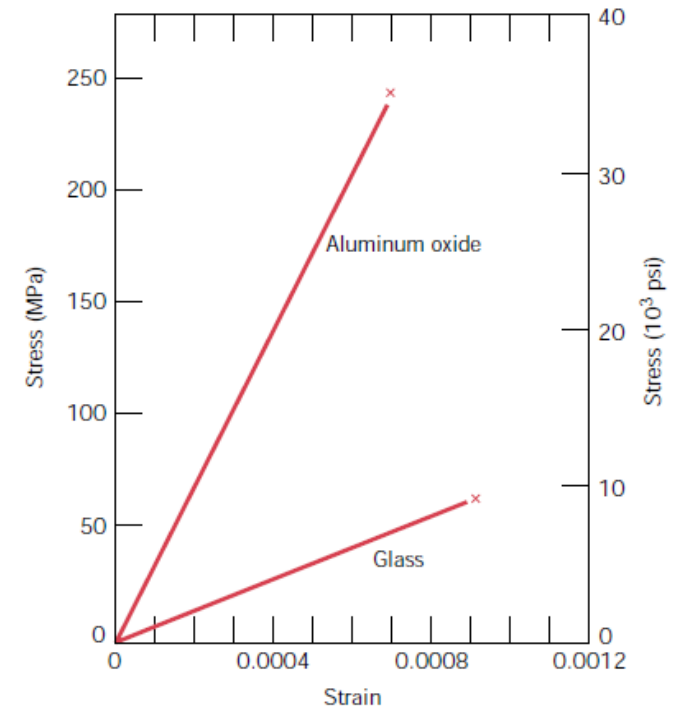
(Al₂O₃, ZrO₂)

Elastic Modulus

- $E \neq f(\text{time the load is applied})$
- $E_{\text{ceramics}} > E_{\text{metals}} \Rightarrow \text{Greater stiffness (bond type)}$
- Low density $\Rightarrow E_{\text{specific}}$ high
- They are employed in the form of fibers in Composite Materials

TABLE 17.1
SPECIFIC MODULI: CERAMICS COMPARED TO METALS

Material	Modulus E (GPa)	Density ρ (Mg m ⁻³)	Specific modulus E/ρ (GPa/Mg m ⁻³)
Steels	210	7.8	27
Al alloys	70	2.7	26
Alumina, Al ₂ O ₃	390	3.9	100
Silica, SiO ₂	69	2.6	27
Cement	45	2.4	19



PROPERTIES OF CERAMICS

Mechanical Properties

Large variability of properties from sample to sample

Reason: porosity that acts as a stress concentrator.

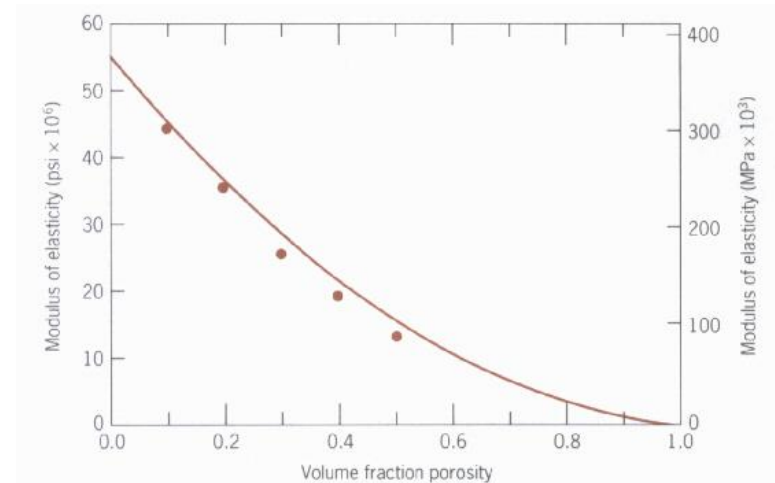
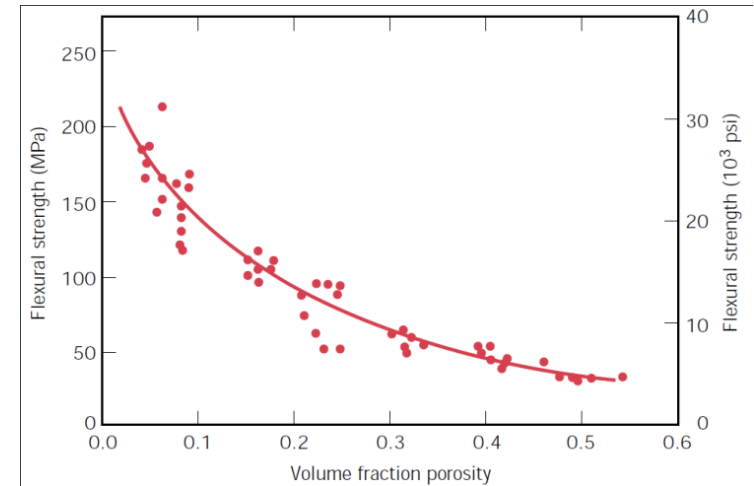
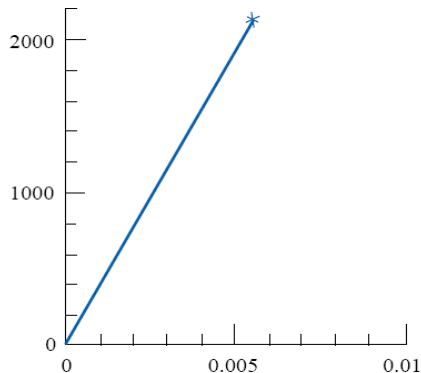
Brittle and catastrophic rupture

Formation of pores during fabrication (bad sintering)

Strength= f (size and volume fraction of pores)

At room temperature ceramic materials break before any plastic deformation occurs:

$$\sigma_y = \text{UTS} = \sigma_{\text{fracture}}$$

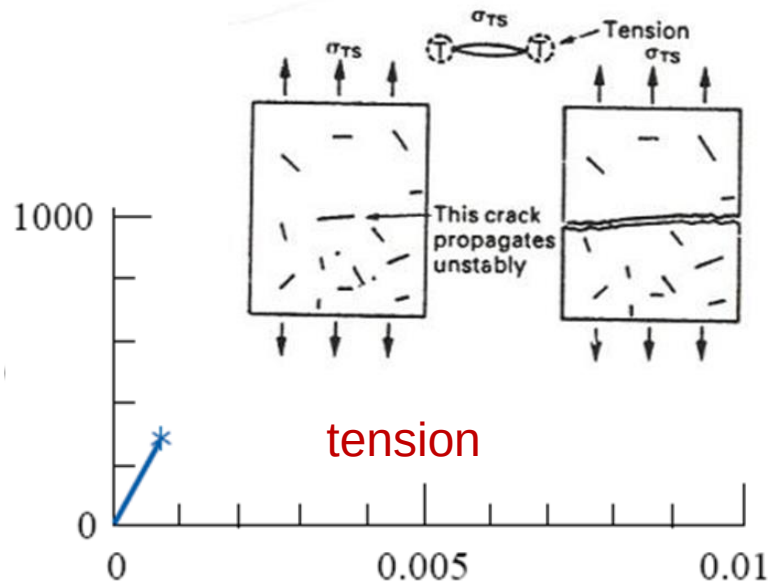


PROPERTIES OF CERAMICS

Testing of mechanical properties

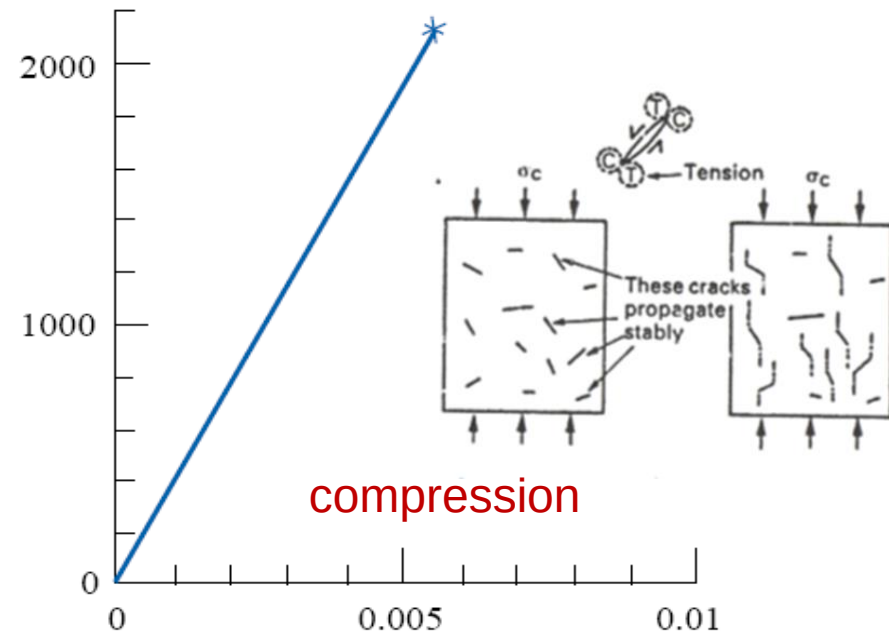
• Tension test

- Difficult to perform tensile test in ceramic material
- Gripping of specimens by clamps produces **cracks**
- Cracks **propagate**



• Compression test

- Simpler than tension tests
- Smaller concentration of stresses in the defects
- Smaller dispersion of results



PROPERTIES OF CERAMICS

Testing of mechanical properties

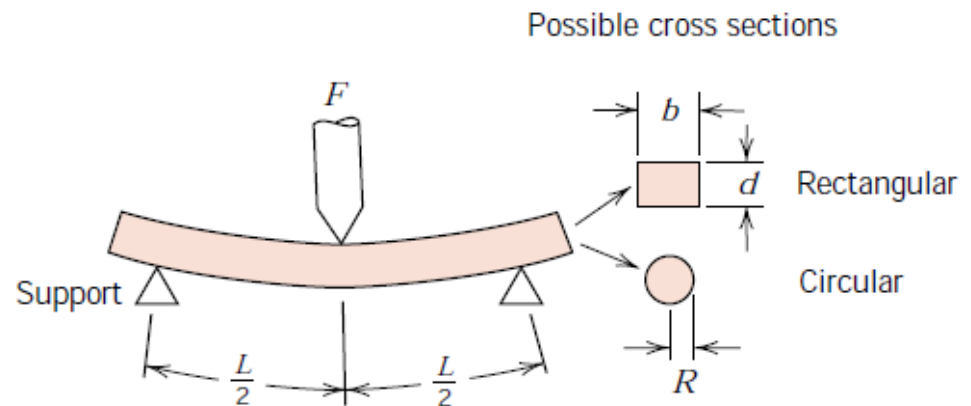
- Bending test**

- 3 point bending test
- At the point where the load is applied the top surface is submitted to compression and the bottom surface is submitted to tension

Material	MOR (MPa)
SiO ₂ glass	70
MgO	200
ZrO ₂	500-1000
Si ₃ N ₄	500-1000

Flexural Strength or Modulus of Rupture

$$\sigma_{fs} = \frac{3F_f L}{2bd} = \frac{F_f L}{\pi R^3}$$

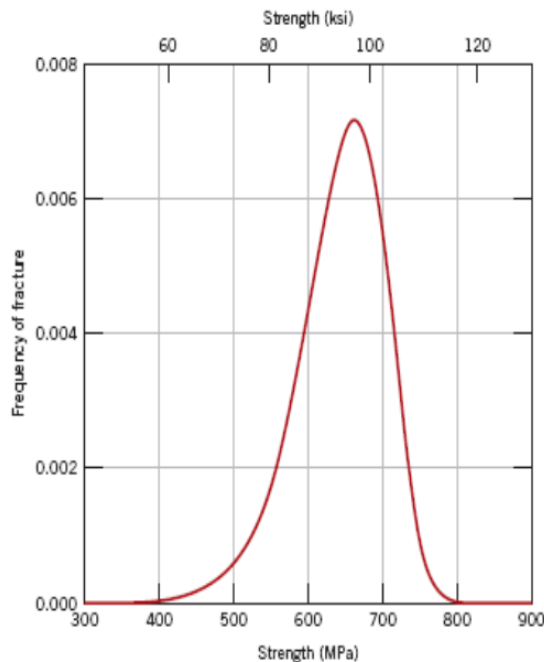


PROPERTIES OF CERAMICS

Testing of mechanical properties

- For ceramic materials, the UTS or σ_{fracture} may **vary** from specimen to specimen, even under **identical testing conditions**
- Dispersion of the results can be described by Weibull distribution

$$f(x; x_0, \lambda, k) = \begin{cases} \frac{k}{\lambda} \left(\frac{x-x_0}{\lambda} \right)^{k-1} e^{-((x-x_0)/\lambda)^k} & x \geq x_0, \\ 0 & x < x_0, \end{cases}$$



k-Weibull modulus

- lower k=greater variation
- ceramics=materials with low k

PROPERTIES OF CERAMICS

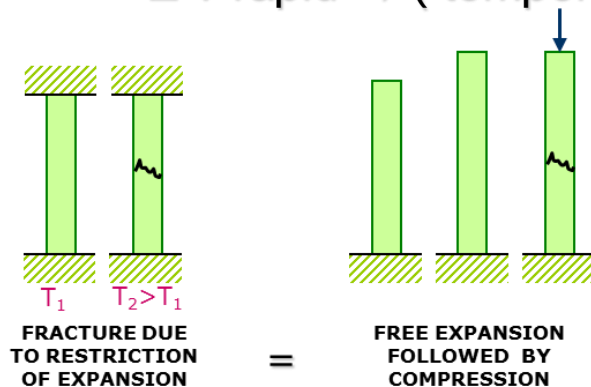
Thermal Shock

Partial or total Fracture due to changes in T

It can be produced by :

Restriction during expansion

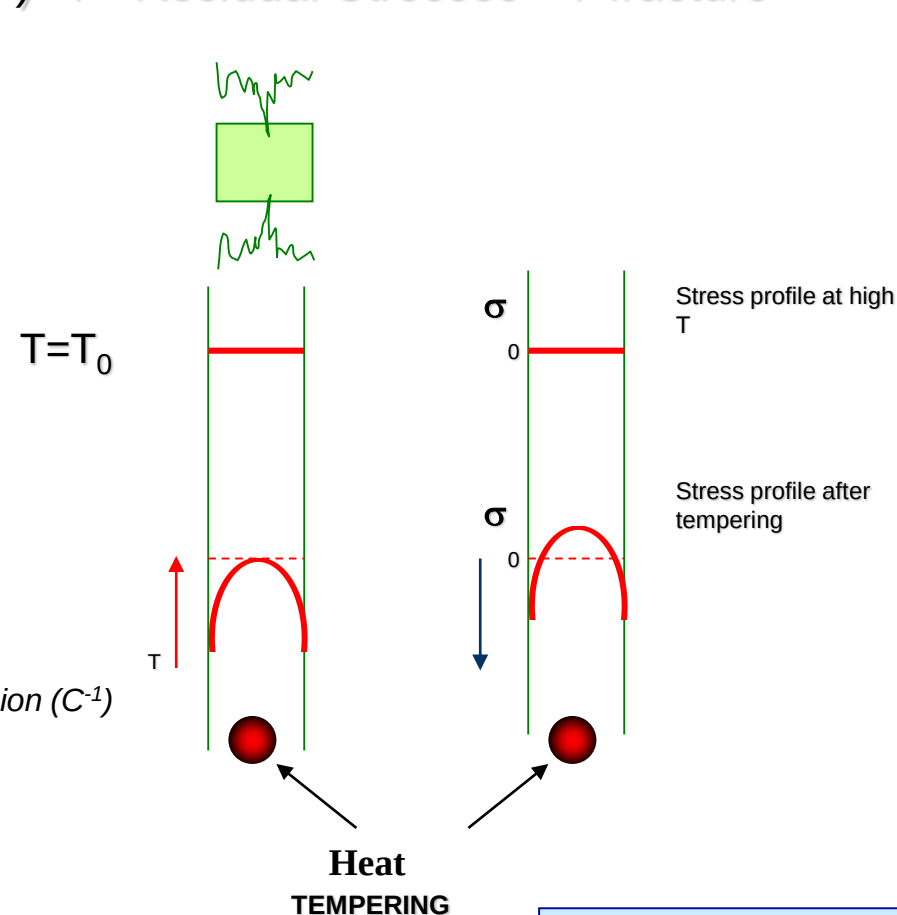
ΔT rapid $\Rightarrow$ (temporary T gradients) $\Rightarrow$ Residual Stresses $\Rightarrow$ fracture



$$\sigma = E\alpha(T_0 - T_f) = E\alpha\Delta T$$

$$TSR = \frac{\sigma_f \cdot k}{E \cdot \alpha_l}$$

E = Youngs Modulus (MPa)
 α = linear coefficient of thermal expansion ($^{\circ}\text{C}^{-1}$)
 σ = thermal stresses (MPa)
 σ_f = Fracture stress (MPa)
 k = Thermal conductivity (W/(m·K))
 ΔT = thermal gradient



PROPERTIES OF CERAMICS

ELECTRICAL PROPERTIES

Ceramic materials are (usually) insulators with high dielectric strength

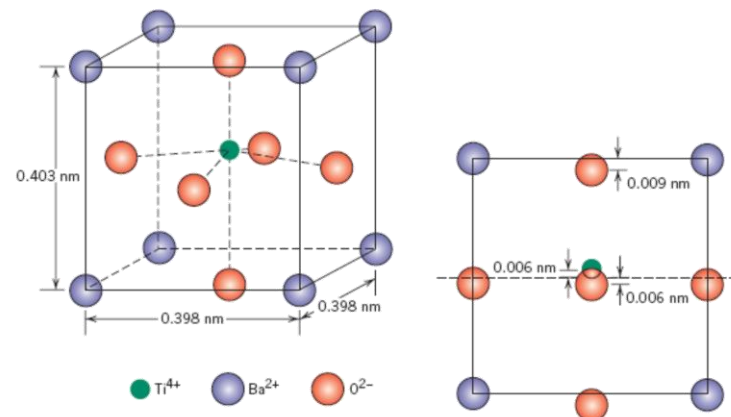
Some ceramics like BaTiO_3 or SrTiO_3 are **ferroelectric**

Ferroelectric compounds have unit cells **without a centre of symmetry**, that results in a **small electric dipole** in the cell.

Ferroelectric compounds exhibit **piezoelectric effect**:

Direct: Compressive stress applied → reduce the distance between dipoles
→ cause a difference in electric potential

Reverse: Voltage is applied → change of its volume



STRUCTURE OF SILICATES

Si and O are the most abundant elements in the earth's crust

They are the base of traditional ceramics

Useful engineering materials because

- Low price
- Great availability
- Special properties

Fundamentally in:

- Construction (bricks, cement, glass)
- Electrical and thermal insulating materials

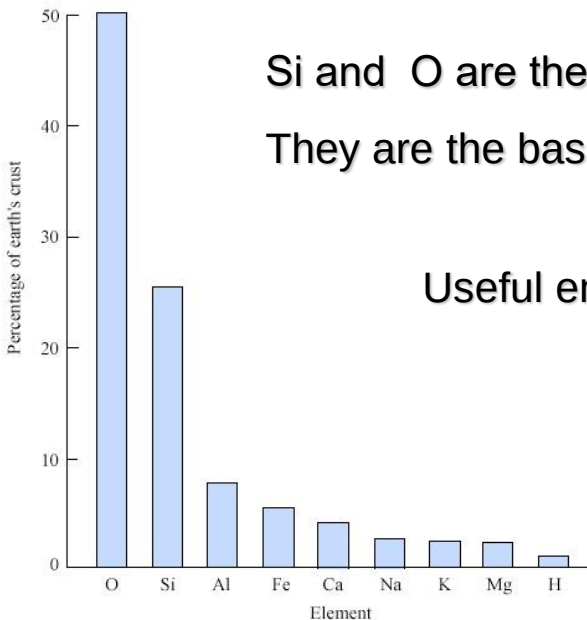
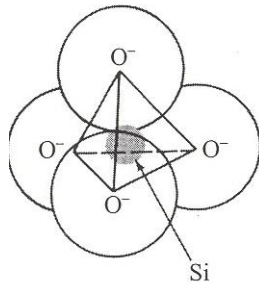


TABLE 8.1 COMPOSITIONS^a OF SOME SILICATE CERAMICS

Ceramic	Composition (wt %)					
	SiO ₂	Al ₂ O ₃	K ₂ O	MgO	CaO	Others
Silica refractory	96					4
Fireclay refractory	50–70	45–25				5
Mullite refractory	28	72				—
Electrical porcelain	61	32	6			1
Steatite porcelain	64	5		30		1
Portland cement	25	9			64	2

^aThese are approximate compositions, indicating primary components. Impurity levels can vary significantly from product to product.

STRUCTURE OF SILICATES

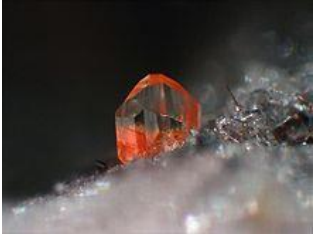


Basic structural unit
of silicates: tetrahedron
with Si in the center
and O in the corners!!

- ⇒ Bond type (*Pauling*): 50% ionic -50% covalent
- ⇒ $r_c/r_A = 0.29 \rightarrow$ stable structure with tetrahedral coordination .
- ⇒ $\uparrow$ packing factor $\Rightarrow$ tetrahedra united in the corners .
- ⇒ Multitude of possible structures: Different structures based on **the same motif**.
 - a) Structures of isolated silicates
 - b) Ring and Chain structures
 - c) Laminar structures
 - d) 3D structures

STRUCTURE OF SILICATES

Classification of silicates as a function of the tetrahedra ordering $[\text{SiO}_4]^{4-}$.



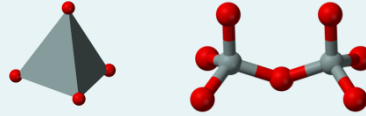
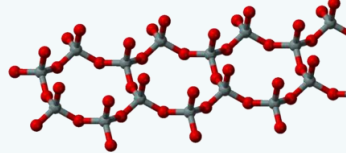
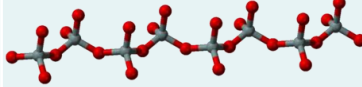
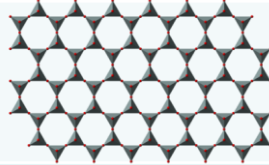
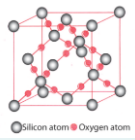
Forsterite (Mg_2SiO_4)



Enstatite (MgSiO_3)



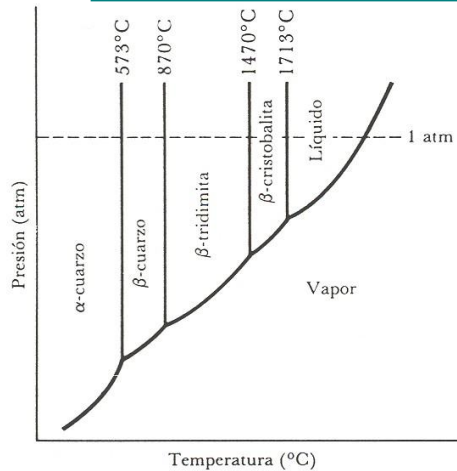
Kaolinite

Type	Structure	Example
island		Orthosilicates & pyrosilicates (MgSiO_4)
ring		Metasilicates - Wollastonite (CaSiO_3)
chain		Enstatite (MgSiO_3)
sheet		Kaolinite $\text{Al}_2(\text{OH})_4^{2+}$ Talc: $\text{Mg}_3(\text{OH})_4^{2+}$
3D		Quartz (SiO_2)

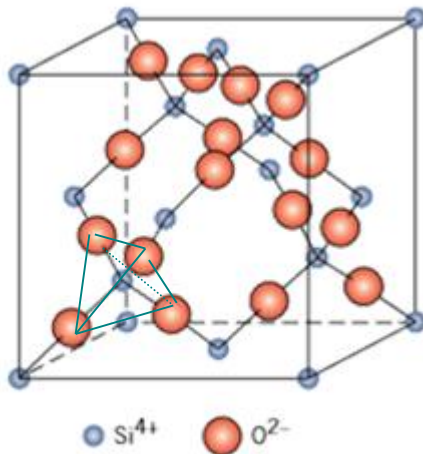
STRUCTURE OF SILICATES

Three-Dimensional Silicates

Silica



- They share all the corners in the tetrahedra
- Unit formula : SiO_2
- Presents Allotropy
- Important component in many traditional ceramics and many types of glasses



Feldspars

- Similar structure to Silica (Al^{3+} replaces Si^{4+}) $\Rightarrow$ lattice with (-) charge $\Rightarrow$ compensates the charge with voluminous cations (Na^+ , K^+ , Ca^{2+} , Ba^{2+}) in interstitial positions .
- Main component of traditional ceramics



AMORPHOUS MATERIALS AND SOLIDIFICATION

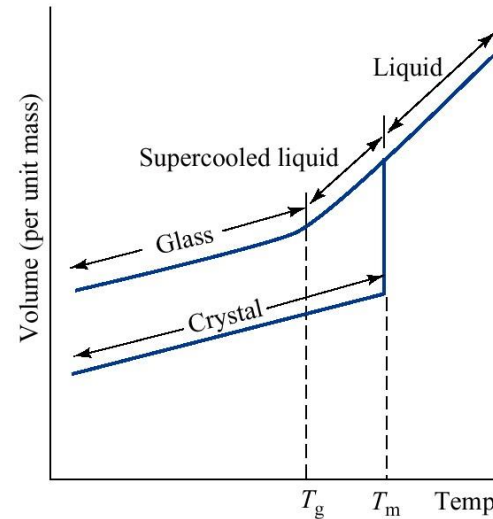
Crystalline Solid

Starting from liquid state, lowering the temperature, solidifies at the melting/solidification temperature (T_m)

GLASS

Starting from the liquid state, lowering the temperature, viscosity increases.

Plastic State $\Leftrightarrow$ Rigid State



Contraction
due to
solidification

GLASS: An **amorphous** inorganic solid with the structure of a liquid. Glasses **harden and become rigid** with a decrease in temperature **without crystallization**

❑ No distinct melting or freezing point

T_g : Glass Transition Temperature: Temperature at which glass phase **transforms from liquid** (or rubber or molten) state **into rigid** or glass state. Associated with amorphous zones

T_m : Melting Temperature. Associated with crystalline zones

$T < T_g \Leftrightarrow$ Glass state

$T > T_g \Leftrightarrow$ Rubbery or leathery state

CONSTITUENTS OF GLASSES

Constituents of (silicate) glasses

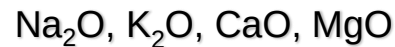
Glass **Formers**



50-99.5 % of the glass material

- exhibit rigid-liquid transition
- **SiO₂**: Fundamental sub-unit : **SiO₄⁴⁻ tetrahedra**
- **B₂O₃**: Fundamental sub-unit: flat triangles BO₃³⁻.
- *Silicate glasses*: former SiO₂
- *borosilicate glasses*: former B₂O₃ + SiO₂.

Glass **modifiers**



3-20 % of the glass material

- oxides that **break the silicate lattice** => lower T_g of glass=>processing at lower temperatures
- ↓ **viscosity** ⇨ facilitate moulding and workability
- They are accommodated in interstitials (do not form part of the silicate lattice)

Glass **Intermediates**



1-30 % of the glass material

Do NOT form glasses only by themselves but can replace silica in the network.

Introduce special properties

- **Al₂O₃** ⇨ tetrahedra AlO₄⁴⁻ replace SiO₄⁴⁻ ⇨ produces **charge defects** (since Al³⁺ and Si⁴⁺). ⇨ ↑ strength at high T (**aluminosilicate glasses**)
- **PbO**
 - Modifies optical properties
 - ↓ T_f (glass soldering)
 - Radiation protection of ↑ E

❑ All glasses are at least 50% silica (SiO₂).
Oxides are added to modify their properties

TYPES OF GLASSES

Soda-lime glass (90% of total glass production) :

- 60-75% silica, 12-18% Na_2CO_3 (soda), and 5-12% CaO (lime)
- The modifiers decrease the glass transition temperature from 1600°C to 730°C

Borosilicate glass

- silicate glass with at least 5% of boron oxide
- boron oxide increases the glass transition temperature
- Label: Pyrex glass (high T applications: ovenware, laboratory equipment etc.)

Lead glasses

- Low T_g - useful for soldering other glasses.
- Shield from radiation.

Glass Type	Composition (wt%)						Characteristics and Applications
	SiO_2	Na_2O	CaO	Al_2O_3	B_2O_3	Other	
Fused silica	>99.5						High melting temperature, very low coefficient of expansion (thermally shock resistant)
96% Silica (Vycor)	96				4		Thermally shock and chemically resistant—laboratory ware
Borosilicate (Pyrex)	81	3.5		2.5	13		Thermally shock and chemically resistant—ovenware
Container (soda-lime)	74	16	5	1		4 MgO	Low melting temperature, easily worked, also durable
Fiberglass	55		16	15	10	4 MgO	Easily drawn into fibers—glass-resin composites
Optical flint	54	1				37 PbO, 8 K ₂ O	High density and high index of refraction—optical lenses
Glass-ceramic (Pyroceram)	43.5	14		30	5.5	6.5 TiO ₂ , 0.5 As ₂ O ₃	Easily fabricated; strong; resists thermal shock—ovenware

CONSTITUENTS OF GLASSES

Optical properties of silicate glasses

- ❑ Silicate glasses are **transparent**. The transparency range depends on composition of the glass
- ❑ "Normal" glass is transparent in the visible part of the EM spectrum.
- ❑ Color of glass is due to presence of dopants.
- ❑ Different ions provide different colors.

COLOURS THAT METALLIC IONS GIVE TO GLASSES		
ION	M ⁺ as a MODIFIER	
	C. I.	COLOUR
Cr ³⁺	6	Blue
Cr ⁶⁺	6	Green
Cu ²⁺	8	Blue –green
Cu ⁺	6-8	Transparent
Co ²⁺	6-8	Rose
Ni ²⁺	8	Yellow –green
Mn ²⁺	6	Light orange
Mn ³⁺	6-8	
Fe ²⁺	6	Blue-green
Fe ³⁺	6-10	Light yellow
U ⁶⁺	6	Light yellow
V ³⁺	6	Green
V ⁴⁺		Blue

PROPERTIES OF GLASSES

Mechanical Properties

- Brittle Materials
- Strength greatly influenced by presence of defects and flaws
- Young modulus depends on the composition, surface imperfections, volume of the material and temperature
- Low Weibull modulus
- Low mechanical strength

Other Properties

Resistant to chemical attack

Electrical Properties

- Generally **insulators** ($\sigma \approx 10^{-10} - 10^{-20} \Omega\text{cm}^{-1}$)
 - High electrical resistivity high dielectric strength
- Conductivity increases with and increase in temperature and the concentration of modifiers

Thermal Properties

Low thermal conductivity
 High thermal expansion coefficient
 ⇒ **Low** thermal shock resistance

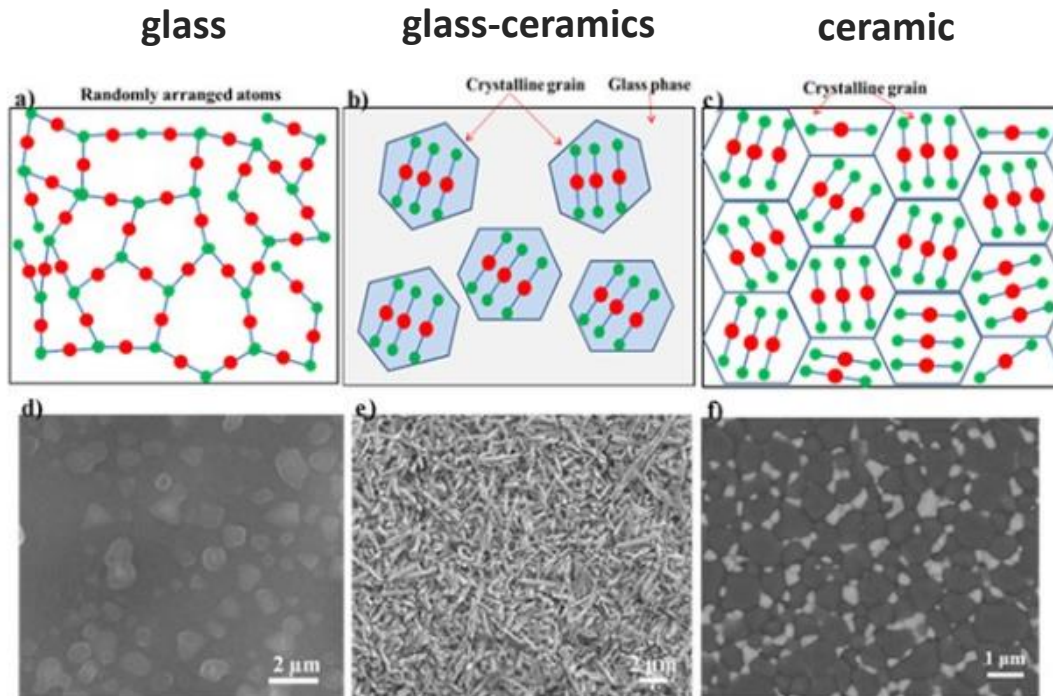
Table ST1 Thermal Shock Resistance of Common Glasses and Glass Ceramics

Material	Thermal Expansion Coefficient ($^{\circ}\text{C}^{-1}$)	Thermal Shock to Failure ^{1*}
Soda-lime-silicate	10^{-5}	80 $^{\circ}\text{C}$
Sodium borosilicate (Pyrex TM type)	3×10^{-6}	270 $^{\circ}\text{C}$
Fused silica	0.5×10^{-6}	1600 $^{\circ}\text{C}$
Lithia-alumina-silicate glass ceramic (Pyrocera TM type)	1.2×10^{-6}	670 $^{\circ}\text{C}$
Transparent lithia-alumina-silicate glass ceramic (Visions TM type)	0.6×10^{-6}	1330 $^{\circ}\text{C}$

^{1*} Assuming failure strength in tension at a stress of 69 MPa.

GLASS - CERAMIC

Glass-ceramics are **polycrystalline** materials produced through controlled crystallization (devitrification) of base glass, producing a fine uniform **dispersion of crystals** throughout the bulk material



- thermomechanical durability $\Rightarrow$ can sustain repeated and quick temperature changes up to 800–1000 $^{\circ}\text{C}$

Common Commercial Composition: $\text{Li}_2\text{O} \times \text{Al}_2\text{O}_3 \times n\text{SiO}_2$ system (**LAS** system).